

DIGITAL ASSIGNMENT-1

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**Course Name: EXPLORATORY DATA
ANALYSIS**

Course Code: BACSE301

Slot: L41+L42

BASIC PYTHON PROGRAMS

QUES 1) Find the factorial of a user input number.

CODE:

```
#taking factorial

n=int(input("Enter number for calculation of factorial "))

prod=1

for i in range(n,1,-1):

    prod*=i

print("The factorial of",n,"is",prod)

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter number for calculation of factorial 7
The factorial of 7 is 5040
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```

QUES 2) Find average of user input list.

CODE:

```
#taking average of list numbers

l1=eval(input("Enter list elements "))

print("The average of elements in the list is",sum(l1)/len(l1))

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter list elements [1,2,6,4,7]
The average of elements in the list is 4.0
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```

QUES 3) Find whether a number is Armstrong or not.

CODE:

```
#armstrong number

n=int(input("Enter the number "))

temp=n

su=0

while(temp>0):

    digit=temp%10

    su+=digit*digit*digit

    temp=temp//10

if(su==n):

    print(n,"is an armstrong number")

else:

    print(n,"is not an armstrong number")

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter the number 153
153 is an armstrong number
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```

QUES 4) Find whether a number is Perfect or not.

CODE:

```
#perfect number

n=int(input("Enter the number "))

su=0

for i in range(1,n):

    if(n%i==0):

        su+=i

if(su==n):

    print(n,"is a perfect number")

else:

    print(n,"is not a perfect number")

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter the number 6
6 is a perfect number
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```

QUES 5) Find whether a number is a strong number or not.

CODE:

```
#strong number

n=int(input("Enter the number "))

def factorial(n):

    prod=1

    for i in range(n,1,-1):

        prod*=i

    return prod

temp=n

su=0

while(temp>0):

    digit=temp%10

    su+=factorial(digit)

    temp=temp//10

if(su==n):

    print(n,"is a strong number")

else:

    print(n,"is not a strong number")

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter the number 145
145 is a strong number
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```

QUES 6) Count vowels and consonants in a word

CODE:

```
#counting vowels and consonents

n=input("Enter the word ")

vow=0

length=0

vowels=['a','e','i','o','u']

for i in n:

    for j in vowels:

        if(i==j):

            vow+=1

print(vow,"is the number of vowels in",n)

print(len(n)-vow,"is the number of consonants in",n)

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter the word amazing
3 is the number of vowels in amazing
4 is the number of consonants in amazing
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```

QUES 7) Find sum of natural numbers using Recursion.

CODE:

```
#sum of natural numbers

def sumofnat(n):

    if(n==0):

        return 0

    else:

        return n+sumofnat(n-1)

n=int(input("Enter upto which number to find sum "))

su=sumofnat(n)

print(su,"is the sum")

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter upto which number to find sum 5
15 is the sum
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```


QUES 8) Find whether a year is a leap year or not.

CODE:

```
#leap year check
year = int(input("Enter a year: "))
if year % 400 == 0:
    print(year,"is a leap year.")
elif year % 100 == 0:
    print(year,"is not a leap year.")
elif year % 4 == 0:
    print(year,"is a leap year.")
else:
    print(year,"is not a leap year.")
print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter a year: 1700
1700 is not a leap year.
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```

QUES 9) Basic Dictionary printing question

CODE:

```
#Dictionary basic program

grades = {

    "Alice": "A",

    "Bob": "B",

    "Charlie": "A+"

}

print("Alice's grade:",grades["Alice"])

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Alice's grade: A
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```

QUES 10) Find the largest of 3 numbers.

CODE:

```
#largest of 3 numbers

a=int(input("Enter first number "))
b=int(input("Enter second number "))
c=int(input("Enter third number "))

if (a >= b) and (a >= c):
    largest = a
elif (b >= a) and (b >= c):
    largest = b
else:
    largest = c

print("The largest number is:",largest)

print("MADE BY PRIYANGSHU DAN 25BAI0014")
```

OUTPUT:

```
● PS C:\Users\priya\Downloads> python -u "c:\Users\priya\Downloads\trial (1).py"
Enter first number 3
Enter second number 4
Enter third number 1
The largest number is: 4
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```